

[Project Title: Identify the Problem and Core Idea]

[Student 1 (UNI)], [Student 2 (UNI)], [Student 3 (UNI)]

COMS 6998: AI-Native Computing | Fall 2026

Project Proposal | Due October 5, 2026, 11:59 PM ET

Track / topic: *[A: Computing for AI, or B: AI for Computing; project topic.]*

Format. Write **two pages**, excluding references. Use the five sections below and replace the guidance and bracketed fields with your project content.

I. PROBLEM AND MOTIVATION

Problem. Describe the problem you want to address in a specific workload, application, or computing system.

Motivation. Explain why it's important and how addressing it could benefit the target application or system.

Research question. State the main question your project will explore and what you aim to achieve.

II. RELATED WORK

Briefly discuss a few relevant papers or systems and explain how your project builds on or differs from them. You can highlight an opportunity for improvement or a question worth exploring. Use citations such as [1]; replace this example with your own references.

III. PROPOSED APPROACH

Describe your main idea and how you plan to implement it. You can outline the key components, the tools or codebases you will use, and the expected contribution. A diagram can help explain the approach.

IV. EVALUATION PLAN

Outline how you plan to evaluate your idea. You can mention potential workloads or datasets, methods to compare against, and useful metrics such as task quality, latency, or cost. An initial plan is sufficient; experiment details can be refined as the project develops.

V. TIMELINE AND TEAM RESPONSIBILITIES

Use the course dates below to outline your planned work and expected deliverables for each stage.

Date	Planned work and deliverables
Oct 19	<i>[Your plan.]</i>
Nov 2	<i>[Your plan.]</i>
Nov 23	<i>[Your plan.]</i>
Dec 7	<i>[Your plan.]</i>
Dec 18	<i>[Your plan.]</i>

Presentations. Midterm presentation: Nov 6. Final poster session: Dec 11.

Team roles. Briefly describe the overall division of work within your team.

REFERENCES

- [1] T. Dao, D. Y. Fu, S. Ermon, A. Rudra, and C. Ré, "FlashAttention: Fast and memory-efficient exact attention with IO-awareness," *arXiv preprint arXiv:2205.14135*, 2022. [Online]. Available: <https://arxiv.org/abs/2205.14135>